WELL-LIFE GENERAL HOSPITAL

Strategic Health Insights Report

Patient Admissions & Diagnosis Analysis | 2021–2024

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Department: Hospital Administration & Clinical Operations

Methodology: FIRE Framework | Tool: Power BI

10,000	44 yrs	7	1,460
Total Patients	Average Age	Diagnoses	Days Analyzed

1. Executive Overview

Between 2021 and 2024, Well-Life General Hospital admitted 10,000 patients across seven distinct diagnoses. On the surface, that number tells a growth story — admissions more than doubled over four years, rising from 1,409 in 2021 to 3,266 in 2024. But the more important story is not the volume. It is what that volume reveals about how the patient population is changing.

When the data is examined across time, the hospital's patient profile has undergone a fundamental shift. Infectious diseases — typhoid and malaria — dominated admissions in 2021 and 2022. By 2024, stroke had become the leading diagnosis, with diabetes and hypertension close behind. The average patient age has risen from 39 to 48 over the same period. The hospital that managed primarily infectious disease four years ago is now managing a population with complex, long-term chronic conditions.

This report applies the FIRE framework — Facts, Insights, Recommendations, and Expected Outcomes — to the four-year patient record dataset. It answers eight specific management questions and delivers a set of strategic recommendations designed to align hospital capacity, staffing, and programming with the reality of the patients now walking through the door.

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2. Business Background

Well-Life General Hospital is a general healthcare provider serving patients across a broad spectrum of medical conditions — from acute infectious diseases to complex chronic conditions requiring long-term management. Its patient population spans all age groups, from adolescents to elderly adults, with male patients accounting for the majority of admissions.

The hospital operates within a Nigerian healthcare context where chronic disease burden is rising, infectious disease remains prevalent in younger populations, and the demand for healthcare services is growing year on year. Management decisions about staffing, bed capacity, specialist recruitment, pharmaceutical procurement, and community outreach all depend on a clear understanding of who the patients are and what they need.

Until this analysis, those decisions were being made without a structured, data-driven view of patient admissions. Records existed — but they had not been connected, modeled, and visualized in a way that made the patterns visible to management. This engagement was commissioned to build that visibility.

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3. The Business Problem

Hospital administration had access to patient records, but lacked a consolidated view of what those records showed in aggregate. Without structured analysis, three specific problems persisted:

First, resource allocation was disconnected from actual patient demand. If chronic disease admissions were rising while infectious disease beds were being maintained at historical levels, the hospital was misaligning capacity with need. The data to correct this existed — but it was not being read.

Second, seasonal planning was reactive rather than proactive. Every year, Q1 produces a spike in admissions across most conditions. Without a visible pattern, the hospital could not prepare staffing, pharmaceuticals, or bed availability in advance. Each spike arrived as a surprise.

Third, demographic shifts were not being captured in service design. As the average patient age rose from 39 to 48 over four years, the services most in demand — specialist chronic care, follow-up management, geriatric support — were changing. But without a time-series view of patient demographics, management had no mechanism to detect or respond to this shift.

Stage 2 of the brief added a data modeling challenge: patient information was stored across separate files — patient records, diagnosis references, and calendar data. Without properly relating these files, any analysis risked inaccurate counts, duplicated records, or missing context. Building a clean, connected data model in Power BI was therefore a prerequisite for all subsequent analysis.

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4. Project Objectives

1. Describe patient volume

Establish the total number of patients admitted over the four-year period and track how that volume has changed year on year.

2. Map the disease landscape

Identify which diagnoses account for the highest patient numbers and determine whether the distribution is shifting over time.

3. Profile the patient population

Analyze the age and gender composition of patients across all diagnoses to understand who the hospital is primarily serving.

4. Detect seasonal patterns

Identify whether certain diagnoses peak in specific quarters or months, enabling proactive planning rather than reactive response.

5. Track admission trends over time

Measure year-on-year growth in patient numbers and connect that growth to changes in average patient age and leading diagnoses.

6. Build a reliable data model

Connect patient records, diagnosis data, and calendar information in Power BI using correct relationships to ensure accurate reporting across all dimensions.

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5. Data Description

Property	Detail
Total Patient Records	10,000
Time Period	2021 – 2024 (1,460 days)
Diagnoses Covered	7 — Asthma, Diabetes, Hypertension, Malaria, Stroke, Typhoid, Ulcer
Diagnosis Categories	Chronic (Asthma, Diabetes, Hypertension, Stroke, Ulcer) | Infectious (Malaria, Typhoid)
Data Source	SkillAhead WellLife Hospital Patient Records (.xlsx)
Visualization Tool	Microsoft Power BI
Data Model	Star schema — Patient Records (fact), Diagnosis table, Calendar table

Field Definitions & Business Meaning

Field	Type	Business Meaning
Patient_ID	String	Unique patient identifier. Used as the primary key in the fact table and ensures each admission is counted once, preventing duplication in aggregations.
Age	Integer	Patient age at admission. The primary variable for age group segmentation and trend analysis — rising average age signals a shift toward an older, more complex patient population.
Diagnosis	Categorical	The medical condition for which the patient was admitted. The central analytical dimension — used to compare patient volumes, track trends, and connect to demographic patterns.

Field	Type	Business Meaning
Gender	Categorical	Patient biological sex (Male/Female). Used to identify gender-specific disease patterns and potential under-utilization of services by one gender.
Admission_Date	Date	The date the patient was admitted. Connects to the calendar table in the data model, enabling year, quarter, and month analysis of admission trends.

6. Data Preparation & Modeling

Stage 2 of the business brief required the construction of a proper data model before advanced analysis could begin. The patient data was stored across three separate files — patient records, diagnosis reference data, and calendar information. Without connecting these correctly, any time-based or diagnosis-level analysis risked producing inaccurate results.

Modeling Decision	Action Taken	Why It Was Necessary
Star Schema Design	Patient Records designated as the fact table; Diagnosis and Calendar tables as dimension tables	A star schema separates measurable data (admissions) from descriptive context (diagnosis names, dates), improving query performance and filter accuracy
Calendar Table	A full date table was created covering all days from 2021–2024 with Year, Quarter, and Month columns	Power BI requires a dedicated date table to enable correct time-intelligence — without it, year-over-year comparisons and quarterly breakdowns can produce errors
Diagnosis Table	A separate diagnosis reference table was linked to the patient records via Diagnosis field	Separating diagnosis metadata from patient records avoids duplication and allows the Chronic/Infectious category filter on the dashboard to work correctly across all visuals
One-to-Many Relationships	Patient Records connected to Calendar (one date, many patients) and Diagnosis (one diagnosis, many patients)	One-to-many relationships ensure filters flow correctly — selecting a year in the calendar table filters all patient records for that year without overcounting
Patient Name Search	A search field was added to enable lookup by patient name directly on the dashboard	Provides clinical staff with quick individual patient access within the same interface used by management for aggregate analysis

7. Management Questions — Answered

Q1 **What is the total number of patients recorded between 2021 and 2024?**

10,000	1,409	3,266	132%
Total Patients	2021 Admissions	2024 Admissions	Growth 2021–2024
2021 – 2024	Starting baseline	Peak year	More than doubled

The hospital recorded exactly 10,000 patient admissions over four years. Growth was not gradual — it accelerated each year. Admissions rose 63% between 2021 and 2022 alone, and continued climbing through 2024. This trajectory signals increasing demand that shows no signs of slowing.

Q2 **What is the total number of patients by diagnosis — and which has the highest and lowest count?**

Rank	Diagnosis	Total Patients	Category	% of Total
#1	Typhoid	1,487	Infectious	14.87%
#2	Asthma	1,440	Chronic	14.40%
#3	Ulcer	1,439	Chronic	14.39%
#4	Malaria	1,436	Infectious	14.36%
#5	Diabetes	1,407	Chronic	14.07%
#6	Hypertension	1,401	Chronic	14.01%
#7 (Lowest)	Stroke	1,390	Chronic	13.90%
Total	All Diagnoses	10,000	—	100%

Typhoid leads overall with 1,487 patients, but the margin between the highest and lowest is only 97 patients across 10,000 total — a remarkably even distribution. This evenness is misleading, however. The distribution within each year tells a very different story: typhoid dominated in 2021–2022, while by 2024, stroke had become the top diagnosis. The aggregate obscures a fundamental shift in the patient mix.

Q4 **What is the average age of patients?**

Diagnosis	Average Patient Age	Primary Age Group	Condition Type
Malaria	27 years	Young adults (21–30)	Infectious
Typhoid	30 years	Young adults (21–30)	Infectious
Asthma	35 years	Working age (31–40)	Chronic
Ulcer	43 years	Middle age (41–50)	Chronic
Diabetes	53 years	Middle-older (51–60)	Chronic
Hypertension	58 years	Older adults (51–60)	Chronic
Stroke	65 years	Elderly (61–70)	Chronic
Overall Average	44 years	Working-middle age	All

The average age pattern by diagnosis is one of the most clinically significant findings in the dataset. Infectious diseases cluster in patients aged 27–30. Chronic conditions cluster in patients aged 35–65. This age gradient means that the hospital's disease profile is directly connected to the age profile of the community it serves — and as the community ages, so will the complexity of care required.

Q5 **How are patients distributed across age ranges?**

Age Range	Total Patients	% of Total	Dominant Conditions
11–20 years	287	2.9%	Malaria, Typhoid
21–30 years	2,023	20.2%	Malaria, Typhoid, Asthma
31–40 years	2,353	23.5% — Largest group	Asthma, Ulcer, Typhoid
41–50 years	1,789	17.9%	Ulcer, Diabetes, Hypertension
51–60 years	1,796	18.0%	Diabetes, Hypertension, Stroke
61–70 years	1,302	13.0%	Stroke, Hypertension, Diabetes
71–80 years	450	4.5%	Stroke, Hypertension

The 31–40 age group is the largest single cohort at 2,353 patients. Combined with the 21–30 group, working-age adults (21–40) account for 4,376 patients — 43.8% of total admissions. This is the hospital's dominant patient segment: employed adults managing both infectious diseases and the early onset of chronic conditions. The 41–80 group accounts for a further 5,337 patients — a population with increasing

complexity of care needs.

Q6 What is the gender distribution of patients?

5,681	4,319	45 yrs	42 yrs
Male Patients	Female Patients	Avg Male Age	Avg Female Age
56.81% of total	43.19% of total	Higher care-seeking threshold	Earlier presentation

Diagnosis	More Prevalent In	Implication
Typhoid	Female patients	Infectious disease exposure may be higher in daily domestic settings
Malaria	Female patients	Potential higher exposure or lower immunity factors
Asthma	Female patients	Environmental and stress-related triggers more frequently reported
Hypertension	Female patients	Female hypertension often linked to stress and dietary patterns
Stroke	Male patients	Men present later with more advanced conditions
Ulcer	Male patients	Lifestyle factors — diet, alcohol, stress — more prevalent in males
Diabetes	Male patients	Consistent with global pattern of higher male diabetes incidence

Males outnumber females in admissions by 1,362 patients. However, the average male patient is 45 — three years older than the average female at 42. This age gap suggests that men are waiting longer before seeking care, arriving with more advanced conditions. The female patient seeking care earlier is a clinical advantage that should be reinforced through female-targeted preventive programs.

Q7 How does patient admission vary by year and quarter — and which year recorded the highest count?

Year	Total Patients	YoY Growth	Top Diagnosis	Avg Patient Age
2021	1,409	Baseline	Typhoid (445)	39 years
2022	2,292	+62.7%	Typhoid (470)	41 years

Year	Total Patients	YoY Growth	Top Diagnosis	Avg Patient Age
2023	3,033	+32.3%	Typhoid (451)	44 years
2024	3,266	+7.7%	Stroke (586)	48 years
Total	10,000	132% over 4 years	Typhoid (overall)	44 years

2024 recorded the highest patient volume at 3,266 — but also the most significant diagnostic shift. Stroke overtook typhoid as the leading condition for the first time. This is not a coincidence: stroke is primarily a condition of older patients, and the average patient age in 2024 was 48 — nine years older than in 2021. The hospital is not simply getting busier. It is getting older.

Quarterly Admission Patterns

Quarter	Pattern	Key Observation
Q1 (Jan–Mar)	Consistent spike across all years	Post-holiday effect — poor diet, deferred checkups, and cold-season stress drive up admissions across most conditions
Q2 (Apr–Jun)	Moderate — second highest	Stabilization after Q1 spike; malaria shows elevation with onset of rainy season
Q3 (Jul–Sep)	Comparable to Q2	Malaria cases peak during rainy season; chronic conditions maintain steady flow
Q4 (Oct–Dec)	Typhoid historically peaks (2021–2023)	Rainy season end and festive gatherings create food hygiene risks driving typhoid — pattern broke in 2024

8. Key Insights

Insight 1: The hospital's patient profile has fundamentally changed

In 2021, typhoid and malaria accounted for the majority of admissions. In 2024, stroke became the leading diagnosis — a condition with an average patient age of 65. This is not a fluctuation. It is a structural demographic shift that requires a structural response in how the hospital is resourced and staffed. A hospital designed for infectious disease management is poorly suited to the patient it will be serving most in five years.

Insight 2: Q1 is predictably the most demanding quarter — and predictability is an advantage

Every year without exception, Q1 produces the highest admission volumes across most conditions. This is not random — it reflects post-holiday dietary patterns, deferred medical visits, and cold-season health stress. A pattern this consistent should be met with preparation, not surprise. The hospital has four years of data confirming that Q1 will be its busiest quarter. That knowledge should be converted into a standing operational protocol.

Insight 3: The 31–40 age group is the highest-demand segment — and the most at risk

Working-age adults aged 31–40 represent the hospital's largest patient group. They are old enough to develop early chronic conditions — diabetes, hypertension, ulcer — while still young enough to dismiss symptoms as stress or overwork. This group is also least likely to access preventive care. The hospital has a significant opportunity to intervene early in this population before conditions progress to the complexity level seen in the 51–70 group.

Insight 4: Male patients arrive later and with more advanced conditions

Men outnumber women in admissions and are three years older on average at presentation. They are disproportionately affected by stroke, diabetes, and ulcer — conditions strongly associated with delayed care-seeking. The implication is that the male patients the hospital is currently treating are not representative of the full male health burden in the community — they are the ones whose conditions became impossible to ignore.

Insight 5: The decline in typhoid and malaria deserves investigation before celebration

Typhoid and malaria volumes decreased between 2023 and 2024, with typhoid losing its usual Q4 seasonal peak entirely. This could reflect genuine public health improvements — better sanitation, mosquito control, community awareness. But it could equally reflect patients choosing other facilities, increased self-medication, or barriers to access. The hospital cannot know which interpretation is correct without direct patient feedback and community health data.

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9. Business Recommendations

Recommendation 1: Expand chronic care capacity — particularly for stroke and diabetes

Stroke is now the leading diagnosis in 2024 with 586 patients. Diabetes and hypertension together account for over 2,800 patients. These conditions require specialist physicians, longer stays, ongoing monitoring, and dedicated outpatient follow-up systems. Current capacity designed around infectious disease management is misaligned with this demand.

> *Expected Impact: 30% increase in chronic care capacity over 12 months. Reduced emergency presentations from unmanaged diabetes and hypertension as more patients access timely outpatient care.*

Recommendation 2: Build a Q1 operational protocol — treat peak season as a known event, not a surprise

Every year, Q1 produces the highest admission volumes. The hospital should treat Q1 preparation as a standing operational routine: pharmaceutical stockpiling in December, roster adjustments in January, and a capacity review in October for the following year's Q1. A quarterly Power BI alert system should flag when any diagnosis approaches its historical Q1 peak.

> *Expected Impact: 25% reduction in medicine shortages and unplanned referrals during Q1. Faster treatment and shorter stays due to better preparation.*

Recommendation 3: Launch targeted programs for the 31–40 age group — the hospital's most valuable intervention point

This is the largest patient segment and the one most likely to develop chronic conditions in the next decade. Workplace wellness partnerships, subsidized screening days, and digital health reminders targeting this demographic would allow the hospital to establish relationships with these patients before their conditions escalate. The cost of early intervention is a fraction of the cost of managing advanced chronic disease.

> *Expected Impact: 25% increase in visits from the 31–40 age group within 18 months. Long-term reduction in chronic disease severity as conditions are detected and managed earlier.*

Recommendation 4: Design gender-specific health campaigns — one strategy cannot serve both populations equally

Female patients are more affected by infectious diseases and present earlier. Male patients are more affected by chronic conditions and present later. These are different problems requiring different responses. Female-focused campaigns should emphasize infectious disease prevention, hygiene, and nutrition. Male-focused campaigns should emphasize chronic disease screening — positioned around convenience: short, fast checkups in workplaces, markets, and transport hubs.

> *Expected Impact: 20% increase in male chronic disease checkups. 30% rise in female attendance for preventive services. Better gender balance in planning and care delivery.*

Recommendation 5: Investigate the decline in typhoid and malaria before reallocating those resources

Before repurposing beds, staff, or equipment previously assigned to infectious disease, the hospital should determine whether the decline reflects genuine health improvement or a loss of patients to other providers or self-treatment. A short community survey and an analysis of local health facility data would provide the answer. Acting on the assumption of improvement when patients are simply going elsewhere would be a strategic error.

> *Expected Impact: A clear evidence base for future infectious disease resource decisions. If the decline is genuine, safe reallocation of resources. If it reflects patient loss, a retention strategy can be developed.*

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10. Expected Outcomes

Strategic Action	Expected Outcome
Expand chronic care capacity	30% increase in chronic care capacity. Reduced emergency cases from unmanaged diabetes, hypertension, and stroke.
Q1 operational protocol	25% fewer medicine shortages. Faster treatment response during the year's busiest period.
31–40 age group programs	25% increase in visits from the largest patient segment. Long-term reduction in advanced chronic disease presentations.
Gender-specific campaigns	20% more male chronic screenings. 30% rise in female preventive attendance. Better care planning for both populations.
Typhoid/malaria investigation	Evidence-based resource reallocation decisions. Either validated savings or a retention strategy — not an assumption.
Power BI dashboard monitoring	Real-time visibility into patient trends. Management making decisions based on current data rather than historical assumptions.

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11. Dashboard Design

A four-page Power BI dashboard was built following the two-stage brief — first building the data model, then layering the analysis and visualizations on top. The dashboard is designed to serve two audiences simultaneously: hospital management reviewing aggregate performance, and clinical staff searching for individual patient records.

Page	Purpose	Key Visuals
Model	Data relationship view showing the star schema connecting patient records, diagnosis table, and calendar table	Relationship diagram with one-to-many connections and filter direction indicators
Analysis	Detailed tabular analysis enabling drill-down by diagnosis, year, and quarter	Year/quarter breakdown tables; quarterly patient counts by condition; growth trend analysis
Visual	Exploratory visual layer for pattern identification across demographics and time	Age group distribution charts; gender split; time-series trend lines for Chronic vs Infectious categories
Dashboard	Executive-facing summary page answering all eight management questions in a single view	KPI cards (10,000 patients, 44 avg age, 7 diagnoses, 1,460 days); diagnosis filter sidebar; total patients by year/quarter/month line chart; patients by diagnosis bar chart; donut chart by gender; age range bar chart; year filter buttons (2021–2024)

12. Limitations

• The dataset contains only five fields — Patient ID, Age, Diagnosis, Gender, and Admission Date. Variables such as treatment outcome, discharge status, length of stay, cost of treatment, referring physician, and ward type are not available for analysis.

• This is a descriptive analysis only — it answers what happened, not why. Causal explanations for trends such as the rise in chronic disease or the decline in typhoid require additional data sources such as community health surveys, environmental data, or referral records.

• The analysis cannot distinguish between new patients and returning patients. A patient admitted three times in 2024 appears as three separate records — meaning total patient volume may overstate unique individuals.

• No financial data is included. The business impact of rising chronic disease admissions — in terms of treatment cost, resource utilization, or revenue — cannot be quantified from this dataset alone.

• The data model connects three tables, but the quality and completeness of the Diagnosis and Calendar tables are assumed. Any inconsistencies in those reference datasets would propagate through all aggregations.

13. Future Improvements

Integrate treatment outcome data

Connecting diagnosis with treatment success, length of stay, and discharge status would transform this from a volume analysis into a quality-of-care analysis — measuring not just who came in but whether they improved.

Add financial data to quantify the cost of the chronic disease shift

Linking patient records to treatment cost data would allow the hospital to see whether rising chronic disease volumes are translating into higher or lower revenue per patient — a critical input for strategic planning.

Build predictive admission forecasts

Using four years of quarterly admission data, a time-series forecasting model could project likely admission volumes for 2025 and beyond by diagnosis — enabling proactive, data-driven capacity planning.

Incorporate community health context

Adding external data such as local population demographics, disease prevalence rates, or socioeconomic indicators would allow the hospital to understand what share of the community's total health burden it is capturing — and where the gaps are.

Patient journey tracking

A unique patient identifier that persists across multiple visits would enable the hospital to track how patient conditions evolve over time — identifying, for example, patients who moved from an infectious disease presentation to a chronic condition in subsequent years.

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14. Technical Skills Demonstrated

Skill	Application
Power Query (M Language)	Imported and transformed patient records from Excel, extracted Year and Quarter from Admission Date, and validated data integrity across all 10,000 records before loading into the model.
Data Modeling (Power BI)	Built a star schema connecting Patient Records (fact table) to a Calendar table and Diagnosis table using one-to-many relationships with correct filter direction — enabling accurate cross-table analysis.

Skill	Application
DAX Measures	Created Total Patients, Average Age, Diagnosis Count, and Days Count measures. Built Chronic vs Infectious category aggregations for the trend line toggle on the dashboard.
Dashboard Design (Power BI)	Designed a four-page dashboard with a diagnosis filter sidebar, patient name search field, year-filter buttons, and six distinct visual types — serving both management and clinical staff audiences.
FIRE Framework Application	Structured all analytical output using Facts → Insights → Recommendations → Expected Outcomes — ensuring each finding connects directly to a business action and a measurable result.
Business Translation	Converted a 10,000-row patient record into a strategic narrative about demographic shifts, seasonal patterns, and intervention priorities — communicating clearly to non-technical hospital management.

15. Conclusion

Four years of patient records tell a story that management could not see without structured analysis. Well-Life General Hospital is growing — but more importantly, it is changing. The patients arriving in 2024 are older, more chronically ill, and more complex to care for than the patients who arrived in 2021. Typhoid, once the leading diagnosis, has been displaced by stroke. The average patient age has risen nine years in four years.

These shifts are not surprising given broader patterns of chronic disease rise in Nigeria — but they are significant. A hospital that continues to allocate resources based on its 2021 patient profile will find itself increasingly misaligned with the patients it is actually serving. The capacity, staffing mix, and community programs that were appropriate four years ago need to be re-evaluated in light of what the data shows.

The Power BI dashboard built in this engagement gives management the tool to monitor these changes in real time going forward. But the most important output is the clarity that now exists about the direction of travel. Well-Life General Hospital knows what is happening to its patient population. It now has both the evidence and the recommendations to respond.

Sign-Off

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Client: Well-Life General Hospital

Department: Hospital Administration & Clinical Operations

Tool: Power BI | Dataset: 10,000 Patient Records (2021–2024)

Data does not replace clinical judgment. It informs it.